

The HDD is a spirited fighter

Although SSDs are quite promising and seem all poised to wipe HDDs off the scene, the latter is not prepared to give up quite easily. There are a lot of exciting developments in this space, which are improving the performance while also bringing down the cost of HDD based storage to really attractive levels...

- Perpendicular magnetic recording technology arranges data bits in an upright and thinner orientation
- Helium-filled drives reduce friction and allow more platters to be squeezed more tightly together
- Shingled magnetic recording technology squeezes hard drive tracks closer together in a way that they overlap each other like shingles on a roof
- Technologies like bit-patterned media recording (BPMR) and heat-assisted magnetic recording (HAMR) are likely to result in 10-terabit-per-square-inch (Tbps) areal densities by 2025
- Seagate has brought to the market a 2TB HDD that is only 7-millimetre thick!
- 100TB HDDs are expected in the next decade

five times compared to new generations of flash. However, the company believes XPoint is the future and it is likely to iron out all these issues within the next few months, so we can still hope to see personal computer and data centre products based on this revolutionary tech. In fact, rumours are that Intel's Optane SSDs might make their way into the next MacBook.

Non-volatile memristor memory

Around two years back, Hewlett Packard Co. announced that it was working on a futuristic and nothing short of revolutionary computer called The Machine, which is to be made of specialised computing cores, photonic connections and an extremely dense, energy-efficient memristor memory. The Machine, now being developed by HP spin-off Hewlett Packard Enterprise (HPE), would be able to handle petabyte-scale data sets as if it were child's play.

The memristor memory part is what we hope has caught your attention now. Memristor memory is a non-volatile memory that uses memristors or electrical components that regulate the flow of current in a circuit and remembers the amount of charge that has previously flowed through it.

If Intel bets the future of SSDs on 3D XPoint, HPE has its hopes for memristor memory. HPE and Sandisk have now partnered to develop a new storage class memory technology, which will combine the best of HPE's memristor project and Sandisk's non-volatile resistive random access memory (ReRAM) technology.

ReRAM, too, is a memristor based technology.

Rough sketches show that the design will involve tiny conductive filaments crisscrossing and connecting silicon layers to represent a bit of data. The resulting non-volatile memory technology is expected to be up to thousand times faster and thousand times more durable than current flash technology. (This is exactly what Intel says about 3D XPoint, so we can well assume that this is how SSDs of the future will be!) It will be byte-addressable like DRAM and will allow systems to employ several terabytes of storage class memory cache per server node for large-scale applications. The resulting technology will be used not only for The Machine but also to make other commercial and enterprise products.

Other companies are also working towards commercialising ReRAM technology by this year. One such is Silicon Valley start-up Crossbar, who hopes that some of their 3D Restive RAM products will be used as memory in wearable devices this year, and in high-density SSDs sometime next year or later.

Looking further into the future

There is a lot of exciting research happening at universities, which hint at what is in store for SSDs well into the future. Here are some examples:

Tantalum based memory. One such research was revealed by scientists of Rice University last year. A team of chemists led by James Tour discovered that applying a voltage to a 250-nanometre-thick sandwich

of graphene, tantalum, nano-porous tantalum oxide (an insulator) and platinum creates addressable bits where the layers meet. Control voltages can be used to shift oxygen ions and vacancies, to switch the bits between ones and zeroes. The team used this to create a solid-state memory technology that allows for high-density 162-gigabit non-volatile storage. The new devices require only two electrodes per circuit, which is simpler than current-generation flash memories.

According to Tour's comments in the university's press reports, "This tantalum memory is based on two-terminal systems, so it's all set for 3D memory stacks... And it doesn't even need diodes or selectors, making it one of the easiest ultra-dense memories to construct. This will be a real competitor for the growing memory demands in high-definition video storage and server arrays."

Further, the tantalum oxide memories can be fabricated at room temperature. The control voltage for write/rewrite is adjustable, allowing a wide range of switching characteristics. The main challenges in the way of commercialising this memory technology are the fabrication of a crossbar device dense enough to address individual bits and a way to control the size of the nano-pores.

BlueDBM. Although not an SSD technology by itself, BlueDBM, an ongoing project at the Massachusetts Institute of Technology has the potential to improve the usage of upcoming flash based storage technologies. A system architecture to accelerate Big Data analytics, BlueDBM comprises a large distributed flash based storage with in-store processing capability and a low-latency high-throughput inter-controller networks.

In some Big Data scenarios, performance is affected by the capacity of fast local DRAM, and in cluster systems with more RAM, the network stack ends up being a bottleneck. BlueDBM attempts to overcome these